What Matters in Relational-to-Ontology Learning? Data Evidence, Completeness Guard, and Scoring

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***Abstract*—Learning an ontology and its database mapping underpins ontology-based data access and knowledge graphs, yet manual authoring is costly while automated generation faces key-free schemas, large-schema output truncation, and convention-dependent scoring. In this paper, across 25 configurations and 16 large language models (LLMs) on the Burr and RODI benchmarks plus five converted databases, we address these challenges via a dual-route framework with four components: (i) DET-V2, a zero-call rule engine driven by instance profiling; (ii) a selective *completeness guard* re-querying only omitted tables; (iii) CERT, a cross-source Learn-then-Test risk controller; and (iv) an instance-level *extent metric*. On eight held-out databases, quality depends less on model scale (13 whole-schema LLMs span 0.032 F1 on seven schemas; ≤ 0.016 renaming shift) than on completeness and conventions. With declared keys, zero-call DET-V2 beats rule-based baselines Ontop and RDB2Onto ($+0.05$–0.27 F1), exceeds published Gemini 2.5 Pro ten-block F1 (0.72 vs. 0.62), and approaches the best LLM (0.611 vs. 0.640); stripping keys drops rule relation F1 to 0.35 while LLMs retain 0.53–0.60. Under complete DDL, per-table decomposition yields -0.03 to 0.00 F1 at $11.6\times$ input tokens, whereas our guard preserves accuracy (within 0.002) at $1.12\times$ tokens, and extent scoring reveals a 0.37–0.46 direction gap.**

***Index Terms*—Ontology learning, relational databases, large language models, data profiling, risk control, benchmarks, knowledge graph construction.**

I. INTRODUCTION

RELATIONAL databases hold most enterprise and scientific data, yet encode domain semantics only implicitly: a `person` table with a `type` discriminator says nothing directly about authors and reviewers, and an undeclared `client_id` column still represents an object relation. Ontology learning from relational databases recovers this conceptual model (classes, relations, attributes, and hierarchies) *together with the executable mapping* tying every element to tables, columns, and joins, forming the backbone of ontology-based data access (OBDA) [1], virtual knowledge graphs [2], and retrieval-augmented enterprise assistants.

For two decades, deterministic bootstrappers dominated: W3C Direct Mapping [3] turns tables into classes and declared foreign keys into object properties, while RDB2Onto-style systems [4] recognise link tables and inheritance. Recently, large language models (LLMs) reshaped the field: given DDL and sample rows, frontier models emit complete ontologies and mappings in one shot, and Burr [5] reports Gemini 2.5 Pro outperforming Ontop and RDB2Onto on its micro-benchmark, while RIGOR [6] decomposes generation table by table. Yet Burr itself carries a warning: on its largest real schema (NPD, 70 tables), Gemini 2.5 Pro drops to 0.30 class F1 against 0.85 for Ontop and 0.70 for RDB2Onto on the constraint-corrected release [7]. To move beyond ad-hoc prompting, we build a unified dual-route framework and investigate four questions on real multi-table databases:

- **RQ1** (*data evidence vs. LLMs*): how far can a zero-call rule engine driven by instance-level data profiling go against published rule and LLM baselines, and where do undeclared keys make LLMs indispensable?
- **RQ2** (*model scaling, decomposition, and completeness guarding*): does quality depend on model scale or per-table decomposition, or can lightweight completeness guarding match whole-schema accuracy at near-single-call cost?
- **RQ3** (*convention transfer and risk control*): do convention-tuned prompts and finite-sample false-discovery bounds transfer from micro-benchmarks to real databases?
- **RQ4** (*extent-based measurement*): what blind spots do syntactic mapping metrics hide compared with instance-level extent evaluation?

Study Protocol: We use the 54 Burr micro-benchmark scenarios as a *development set* for rule, prompt, and risk-control calibration, while sealing Burr's four held-out schemas so only the scoring program opened their gold mappings; all post-run rule updates (DET-V2) and stress tests were pre-registered. After freezing the codebase, we converted and execution-verified five additional OBDA databases (two real, three generated) and evaluated on RODI [8]. A pre-registered extension tested identifier renaming, complete key removal (`@stripped`), and four Claude plus three GPT models, followed by a 2,049-call complete-DDL paired experiment across eight databases.

Key Findings: Across 25 configurations and 16 LLMs, real-database mapping quality is governed far less by model choice than by *input/output completeness* and *gold modelling conventions*: (1) *Data-profiling rules rival frontier LLMs when keys are declared, whereas key-free schemas isolate LLM value.* With zero model calls, DET-V2 outperforms Ontop and RDB2Onto by $+0.05$ to $+0.27$ F1 across Burr benchmarks, surpasses published Gemini 2.5 Pro ten-block F1 (0.72 vs. 0.62), and reaches 0.611 across eight held-out databases—within 0.029 of the best LLM (0.640, Fig. 1). Stripping declared keys drops rule relation F1 from 0.56 to 0.35, whereas LLMs retain 91–100% (0.53–0.60). (2) *Model choice is second-order; selective completeness guarding replaces costly per-table decomposition.* Thirteen whole-schema configura-

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tions span just 0.032 F1 across seven held-out databases (and shift ≤ 0.016 under synonym renaming). An apparent collapse on 70-table NPD stemmed from silent prompt-builder truncation (Section VI-B1) and early output cutoff (27% of runs on 51–100-table schemas across 9,673 outputs). Under complete DDL, full per-table decomposition yields -0.03 to 0.00 F1 at $11.6\times$ input tokens, whereas our *completeness guard* requeries only omitted tables and retains single-call accuracy (within 0.002 F1) at $1.12\times$ tokens (Fig. 6). (3) *Conventions and evaluation semantics govern transfer and rankings.* Micro-benchmark convention prompts drop from 0.95 to 0.50 on real schemas, Learn-then-Test [9] FDP bounds fail under cross-schema convention shifts, and scoring relation direction via instance extents lowers rule-system relation F1 by 0.37–0.46.

Contributions:

1) **Instance-Driven Rule Engine (DET-V2) and Key-Removal Boundary**: A zero-call DuckDB profiling and rule engine over UCCs, INDs, FDs, and value-partition discriminators that substantially beats Ontop and RDB2Onto ($+0.05$ to $+0.27$ F1), matches per-table LLM pipelines (0.611 vs. 0.585–0.619), and demarcates where undeclared keys require LLMs (Sections IV-A and VI-C).
2) **Completeness Census and Selective Completeness Guard**: A diagnosis across 16 LLMs, 9,673 outputs, and a 2,049-call paired experiment showing why full per-table decomposition is cost-ineffective ($11.6\times$ tokens for ≤ 0.00 mean gain), paired with a selective table-re-asking guard that preserves accuracy at $1.12\times$ tokens (Sections IV-B, VI-A and VI-B).
3) **Cross-Source Risk Control (CERT) and Transfer Analysis**: The first Learn-then-Test FDP controller over pooled rule and LLM ontology elements, with a pre-registered diagnosis of why correlated convention errors break micro-benchmark bounds on real schemas while retaining uncertainty ranking (50% vs. 8% precision) (Sections IV-D and VI-D).
4) **Instance-Level Extent Metric and Verified Benchmark Suite**: A tuple-extent Jaccard evaluator exposing 0.37–0.46 direction sensitivity, a standalone Burr scorer with zero disagreements across 640 scenario–system units and 222 paired mappings, and five execution-verified OBDA benchmark mappings (332 entries) (Sections III, V and VI-E).

II. RELATED WORK

Rule-based OBDA bootstrapping: W3C Direct Mapping [3] and R2RML [11] standardised relational-to-RDF mappings. D2RQ [12] and Ontop [1] bootstrap direct mappings with lightweight refinements, while BootOX [13] and RDB2Onto [4] add rules for link tables, inheritance, and denormalisation. Because these systems inspect only declared schema constraints, they fail when constraints are omitted or implicit in data values. Our DET/DET-V2 engine extends this family by grounding role, hierarchy, and split decisions in instance-level data profiling (Section IV-A), outperforming Ontop and RDB2Onto even on constraint-complete schemas while remaining runnable when keys are missing.

LLM-based ontology and mapping generation: LLMs construct taxonomic backbones [14], address text-centric LLMs4OL tasks [15], [16], build ontologies from tabular files via multi-step prompting [17], and emit RML for JSON given a target ontology [18]. On relational databases, LLM4VKG [2] maps tables to a given ontology under RODI queries, whereas RIGOR [6] traverses foreign keys to prompt one fragment per table with retrieved context and a second LLM verifier. On Burr's ISWC schema, RIGOR reports $0.73/0.67/0.52$ class/relation/attribute F1—scores matched by a single whole-schema Kimi-K3 call in our study (Table I).

Novelty over prior work: Burr [5] introduced syntactic mapping equivalence and initial baselines. We advance beyond Burr, RIGOR [6], and LLM4VKG [2] along four axes: (i) an instance-profiling rule engine (DET-V2) that rivals frontier LLMs at zero token cost when keys exist; (ii) a controlled comparison of whole-schema vs. RIGOR-style per-table decomposition plus a selective *completeness guard* that captures truncation resilience at $1/10$ the token overhead; (iii) finite-sample cross-source risk control (CERT) tested across sealed schema distributions; and (iv) an instance-level extent metric and five execution-verified OBDA benchmarks.

LLMs on database structure and context limits: LLMs support schema matching [19], column typing [20], documentation [21], and foreign-key detection [22], [23], though accuracy degrades on enterprise schemas [24]. Both schema matching [25] and text-to-SQL face context trade-offs: Maamari et al. [26] skip schema linking when the full schema fits context, whereas Spider 2.0 [27] and AutoLink [28] navigate $> 1{,}000$-column schemas incrementally. Our census adds two distinct failure modes: silent DDL filtering before the model and output-budget truncation during generation.

Data profiling, risk control, and evaluation: Our evidence layer builds on unique column combinations [29], inclusion dependencies and learned FK scoring [30], [31], [32], functional dependencies [33], [34], [35], learned BI joins [36], and LLM-adjudicated dependencies [37], turning profiled signals into ontology modelling rules. For reliability, Learn-then-Test [9] and risk-controlling prediction sets [38] bound arbitrary risks under exchangeability, and RTS [39] applies conformal abstention to text-to-SQL; CERT is, to our knowledge, the first FDP controller for learned ontologies and exposes why exchangeability fails across schema conventions. For evaluation, RODI [40], [8] tests query answers (conflating ontology learning with vocabulary alignment), Burr [5] compares mapping syntax (ignoring relation direction and property-class attachment), and recent benchmarks study source-to-term mapping [41] or competency questions [42]; our extent metric directly evaluates mapped instance tuples alongside both views.

III. TASK, DATA AND PROTOCOL

Task: Given a relational database $D = (S, I)$ with schema S (tables, columns, types, and any declared primary, foreign, or uniqueness constraints) and instance I, a system outputs a mapping M in Burr's JSON dialect of D2RQ comprising three element types: *class maps* (class

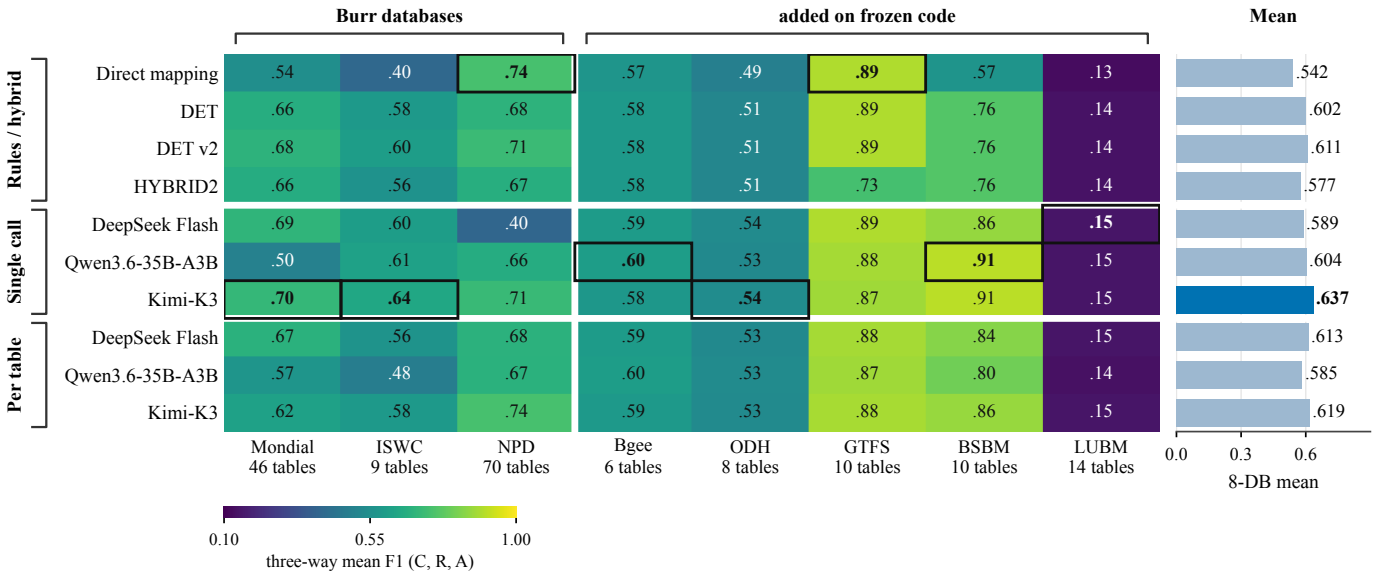


Fig. 1. Three-way mean F1 (class, relation, attribute) for ten systems across eight held-out databases with data (table counts beneath names; per-type values in [10, Table S4]). Left: Burr held-out databases. Right: five databases added after code freeze (Bgee, ODH: real data; GTFS, BSBM, LUBM: generated). Framed cells mark column bests. Zero-call DET-V2 (0.611) beats DM (0.542, +0.069), matches per-table LLM pipelines (0.585–0.619), and sits within 0.026 of whole-schema Kimi-K3 (0.637). Per-table NPD cells reflect the initial prompt builder (Section VI-B1); LUBM is a floor because its gold IRIs combine cross-table values.

name, key-column URI template, optional conditions, joins, and `subClassOf`), *object property maps* (linking two class maps via join conditions), and *data property maps* (linking a property to a column or SQL expression). The ontology is induced by M, whose executable grounding makes every element checkable.

Benchmarks: **Burr** [5] supplies 54 micro-benchmark scenarios across six categories (Basic, Attributes, Normalization, Relationships, Hierarchies, Special) and four held-out schemas: Mondial (46 tables, with foreign keys), ISWC (9 tables), NPD FactPages (70 tables in released DDL, 244,687 rows), and SAP Enterprise (7 wide tables, schema only). Burr equates two class maps when their URI-template columns, conditions, and joins match, and two property maps when their joins, columns, and conditions match; precision and recall count bag duplicates, and `subClassOf` is unscored. Our benchmark copy is byte-identical to the authors' constraint-corrected release [7]; for Mondial and NPD we report corrected Ontop and RDB2Onto scores (‡), while Gemini 2.5 Pro and other baseline figures follow the published paper (†) [10, Sec. S6].

Five Execution-Verified Benchmark Databases: After freezing our code, we converted five human-authored R2RML/Ontop benchmarks into Burr's format: Bgee [43] (gene expression, 6 tables, real data), Open Data Hub [44] (ODH tourism, 8, real data), GTFS-Madrid-Bench [45] (transit, 10), BSBM [46], [47] (e-commerce, 10), and LUBM4OBDA [48] (university, 14). Of 342 source entries, 285 convert exact-equivalently, 47 are approximations (encoding functions or external IRI attributes), and 10 (2.9%) are inexpressible; we retain 332 (gold-vs.-gold self-consistency 1.0). Executed directly on the instance data, 325 of 332 entries return identical tuples to their source mappings (11 up to encoding punctuation) and 7 are empty on both sides. A two-pass audit of 233 entries fixed 14 LUBM entries attached to the wrong class without altering mapping-based scores [10, Sec. S5]. Because these schemas largely model one class per table, systems separate primarily on relations and attributes. We also evaluate on RODI [8] via query answering [10, Sec. S2].

Anti-Leakage Protocol: Because micro-benchmark gold mappings expose specific modelling conventions, we strictly separate data roles. The 54 scenarios serve as the *development set* for rule, prompt, and CERT design. Burr's four held-out schemas remain *sealed*: only the scorer read their gold mappings, and every post-run modification (DET-V2 rules, stress tests) was pre-registered with rationale and expected direction prior to execution. The five added databases and RODI ran once on frozen code, and the extension round (Section VI-A: synonym/opaque renaming, `@stripped` key removal removing all declared PK/FK/unique constraints, and new frontier models) was registered before API calls. "The eight held-out databases" denotes Burr's three data-bearing schemas plus the five added ones.

Evaluated Systems: Each system traverses a path in Fig. 2 (inputs, calls, and NPD costs in [10, Table S14]): W3C Direct Mapping (DM), our data-profiling engines (DET, DET-V2; Section IV-A), whole-schema LLM generation (WHOLE-DB), per-table generation and selective completeness guarding (PER-TABLE; Section IV-B), rule-skeleton hybrid (HYBRID2; Section IV-C), and LTT risk control (CERT; Section IV-D). The main study uses DeepSeek V4.1 Flash (reasoning enabled by default) as reference plus an eight-model ladder from locally quantised Qwen3-14B to Kimi-K3 and Qwen3.8-Max at temperature 0 over K samples; paired complete-DDL settings are in Section VI-B3 [10, Sec. S9].

IV. METHODS

Fig. 2 illustrates our dual-route architecture comprising six modular components between raw relational tables and a verified ontology mapping. The *evidence route* profiles instance data (component 1) and fires deterministic modelling rules (2) at zero LLM cost. The *model route* prompts an LLM once for the whole schema, table by table, or via selective completeness guarding (3) and unifies fragments in dependency order (4). A pooling module (5) either grounds HYBRID2 on the rule skeleton with data-checked LLM refinements or feeds our cross-source agreement risk controller CERT (6), which partitions elements into certified, suggested, and review sets.

A. Evidence Route: Data Profiling and Rules (DET, DET-V2)

Executed entirely in DuckDB [49] within seconds on standard schemas, component 1 profiles column statistics, exact and approximate ($g_3 \leq 1\%$) minimal unique column combinations up to arity three via levelwise Apriori pruning [29], and a ranked identity key preferring declared keys and natural over surrogate columns. Across tables, it mines unary inclusion dependencies (≥ 0.95 coverage to a unique parent), exact functional dependencies (up to 5×10^4 rows), *value partitions* (low-cardinality discriminators inducing systematic null/non-null column blocks), foreign-key cardinalities, link-table structures, and isomorphic tables.

Component 2 applies ablation-ready rules in three phases: *Phase A* resolves primary/identity keys and joins from declared foreign keys plus name-cued inclusion dependencies. *Phase B* assigns table roles: pure binary link tables become object properties over two joins; enriched link tables become classes with two properties; tables keyed by a parent foreign key become subclasses. *Phase C* maps entity tables (key as URI template, non-keys as data properties, foreign keys as object properties), turns value partitions into conditional subclasses, and splits FD-determined surrogate sub-concepts into separate classes.

Following the initial held-out run, we pre-registered four textbook refinements (DET-V2): firing conditional subclasses only on *type-like* discriminators ($\leq$ 8 values, $\geq$ 2 exclusive non-null columns, identifier in `type/kind/category/class/role`); requiring lexical cues for self-reference and FD concept splits; testing parent-to-child relation orientation; and early stopping in uniqueness search. Because the first two never trigger on the development set, they cannot overfit it.

B. Model Route and Selective Completeness Guarding

WHOLE-DB adapts Burr's whole-schema checklist prompt (DDL plus five sample rows per table), emitting D2RQ JSON rather than Turtle (0.74 vs. 0.72 dev F1). Above 60k DDL characters, a compact filter retains `CREATE` and constraint `ALTER` statements (after fixing the comment-stripping defect analysed in Section VI-B1).

PER-TABLE issues one call per table T with all table names, the DDL and five rows of T plus up to eight FK/IND neighbour tables, and an ownership instruction restricting output to T's classes, conditional subclasses, data properties, and child/link object properties. Component 4 merges fragments in dependency order under Burr's equivalence keys $\kappa(\cdot)$ (extended with raw expressions to preserve constant-IRI classes and SQL attributes), letting parent definitions override child stubs and unifying co-keyed class names (with optional *ctx* name propagation), at 4–22$\times$ the input tokens of WHOLE-DB.

To avoid PER-TABLE's token overhead while preventing silent table omissions on large schemas, we introduce a *selective completeness guard*. Let $\mathcal{T}(S)$ be all tables in schema S and $\mathrm{tbl}(M_{\mathrm{WD}}) \subseteq \mathcal{T}(S)$ be the tables referenced by at least one element of a single-call mapping M_{WD}. The guard queries only the omitted set $\mathcal{T}_{\mathrm{miss}} = \mathcal{T}(S) \setminus \mathrm{tbl}(M_{\mathrm{WD}})$ via PER-TABLE and merges fallback fragments *behind* M_{WD}:

$$M_{\mathrm{guard}} = M_{\mathrm{WD}} \cup \left\{ m \in M_{\mathrm{PT}}(\mathcal{T}_{\mathrm{miss}}) \;\middle|\; \kappa(m) \notin \kappa(M_{\mathrm{WD}}) \right\}, \tag{1}$$

where $M_{\mathrm{PT}}(\mathcal{T}_{\mathrm{miss}}) = \bigcup_{T \in \mathcal{T}_{\mathrm{miss}}} M_{\mathrm{PT}}(T)$, so global whole-schema modelling choices win every conflict while truncated or omitted tables are selectively recovered (Section VI-B3).

C. Hybrid Route: Rule Skeleton with Verified Semantics (HYBRID2)

HYBRID2 locks in DET's deterministic keys, table roles, and structural joins as a hallucination-free skeleton, accepting PER-TABLE proposals only for schema-underdetermined semantics (conditional subclasses, concept splits, composite SQL expressions, boolean-flag relations, names, and directions) provided each proposal passes instance checks: existing near-unique template columns, non-empty condition extents, and joins backed by declared keys or ≥ 0.95-coverage INDs.

D. Cross-Source Agreement Risk Control (CERT)

CERT deduplicates pooled elements across R distinct system arms and S_{LLM} total LLM samples under equivalence keys $\kappa(\cdot)$, assigning each unique candidate e a cross-source agreement score:

$$a(e) = n_{\mathrm{arm}}(e) + \frac{1}{2}\,\frac{n_{\mathrm{samp}}(e)}{S_{\mathrm{LLM}}}, \tag{2}$$

where $n_{\mathrm{arm}}(e)$ counts distinct proposing arms and $n_{\mathrm{samp}}(e)$ counts proposing LLM samples. For each element type, Learn-then-Test [9] scans thresholds $\tau_1 > \cdots > \tau_L$ on leave-one-category-out development calibration, testing $H_\tau : \mathbb{E}[\mathrm{FDP}(\mathcal{C}_\tau)] > \alpha$ at the database level via Hoeffding–Bentkus p-values ($\delta = 0.1$) to guarantee $\Pr(\mathbb{E}[\mathrm{FDP}(\mathcal{C}_{\hat{\tau}})] \leq \alpha) \geq 1-\delta$ under database exchangeability, partitioning candidates into *certified*, *suggested*, and *review* queues without touching real-database gold mappings.

V. MEASUREMENT

Verified Standalone Burr Scorer: Burr's official evaluator requires PostgreSQL and an experiment-tracking stack to round-trip JSON mappings through D2RQ Turtle and score bag-based equivalence. We built a dependency-free Python scorer faithfully reproducing the Turtle round trip and both

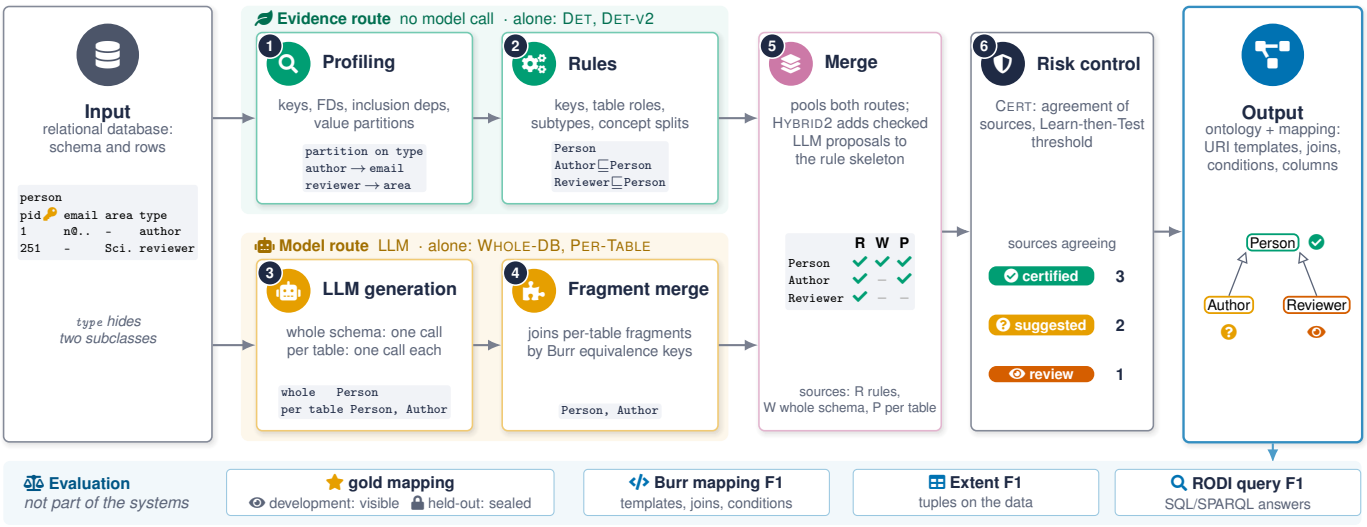


Fig. 2. Dual-route system architecture. Input: relational database (schema and rows); output: ontology with executable table mappings. Profiling (1) and deterministic rules (2) form the zero-call evidence route (DET/DET-V2); whole-schema, per-table, or completeness-guarded LLM generation (3) and dependency-ordered merge (4) form the model route; pooling (5) supports data-checked hybridisation (HYBRID2); and Learn-then-Test risk control (6, CERT) certifies elements by cross-source agreement (illustrated on `hierarchy/denormalized`).

quirks of the original (retaining only the first predicate of compound conditions and leaving `subClassOf` unscored), achieving 1.0 gold self-consistency across all 58 cases. Cross-checked against Burr's official Docker evaluator, our scorer yielded *zero* disagreements across all 640 scenario–system units in the main study and all 222 mappings in the paired experiment.

Instance-Level Extent Metric: Syntactic mapping comparison scores *how* a mapping is written rather than *what* it denotes: equivalent subtypes written via joins vs. conditions fail to match, whereas a relation and its inverse share identical join sets so Burr's metric is blind to relation direction. We therefore introduce an *extent-based metric* that executes every mapping element e on instance data I to materialise its denoted tuple set $[\![e]\!]_I$—key tuples for classes, (subject key, object key) pairs for object properties, and (subject key, literal) pairs for data properties—and matches predicted elements $\hat{M}$ to gold elements $M^\star$ one-to-one via maximum-weight bipartite matching (Hungarian assignment) over Jaccard similarity:

$$J(e,g) = \frac{|[\![e]\!]_I \cap [\![g]\!]_I|}{|[\![e]\!]_I \cup [\![g]\!]_I|}, \qquad J(e,g) \geq \tau \tag{3}$$

at strict threshold $\tau = 1$ (gold self-consistency 1.0 across all 57 data-bearing cases). Extent execution also uncovered 12 empty-extent gold defects (e.g., misspelt or non-existent columns).

Statistical Protocol: We summarise class, relation, and attribute F1 by their unweighted *three-way mean* (category means on Dev; equal-weight database means on held-out sets after averaging K LLM samples). Pairwise comparisons use 10,000-resample paired bootstrap with familywise Holm correction [50]; because four or eight databases afford coarse power, held-out differences are treated descriptively without reading non-significance as equivalence.

VI. RESULTS

Table I compares published baselines, W3C Direct Mapping, and our dual-route configurations across Burr's development set, four sealed held-out schemas, and ten-block summary. Our zero-call engine DET-V2 surpasses the strongest published baselines by $+.09/+.12/+.04$ on Dev, $+.07/+.05/+.06$ on Held-out mean, and $+.11/+.13/+.07$ on Ten-block, while whole-schema Kimi-K3 raises Held-out mean to $.62/.54/.57$ ($+.14/+.10/+.05$ over the best published baseline and $+.21/+.18/+.11$ over published Gemini 2.5 Pro). We examine model scaling and decomposition (**RQ2**), data-profiling rules vs. key removal (**RQ1**), transfer and risk control (**RQ3**), and extent measurement (**RQ4**).

A. *RQ2: Model Scaling Is a Second-Order Factor*

Fig. 3a plots nine models under whole-schema generation with complete prompts. Eight of nine span a narrow 0.72–0.76 three-way F1 band on Dev (locally quantised Qwen3-14B reaches 0.67), and scaling open-weight parameters from 27B to 397B yields flat scores $(0.72, 0.74, 0.74, 0.73)$. Across Burr's four held-out schemas (0.42–0.58), separation tracks output completion rather than model size: the two lowest-scoring models are those most frequently truncated on 70-table NPD.

Adding four Claude and three GPT models in the pre-registered extension round [10, Sec. S4] confirms this invariance (Fig. 3b): across the seven held-out databases excluding NPD, all 13 whole-schema configurations cluster within 0.595–0.627 (vs. 0.596 for zero-call DET-V2). Including NPD, the strongest model (Claude Opus 5.5, 0.640) leads DET-V2 (0.611) by just 0.029 (non-significant), and spread across models is governed almost entirely by NPD completion. For DeepSeek, default API reasoning tokens exhausted the output budget; disabling thinking changes scores by < 0.03 while completing every NPD run.

TABLE I

BURR MAPPING-BASED F1 (CLASS/RELATION/ATTRIBUTE). DEV: SIX-CATEGORY MEAN. HELD-OUT: PER SCHEMA AND FOUR-SCHEMA MEAN. TEN-BLOCK: UNWEIGHTED MEAN OF SIX DEV CATEGORIES AND FOUR HELD-OUT SCHEMAS (BURR'S OWN AVERAGE ALSO INCLUDES FK-FREE MONDIAL). LLM ARMS AVERAGE K SAMPLES (SD $\leq$ .03, EXCEPT WHOLE-DB DEEPSEEK ON NPD WITH TWO INCOMPLETE SAMPLES, SD .08). PER-TABLE AND HYBRID2 ON NPD USE THE INITIAL BUILDER (SECTION VI-B1). †: PUBLISHED VALUES [5]; ‡: CONSTRAINT-CORRECTED ONTOP/RDB2ONTO VALUES [7], [10]. BOLD: COLUMN BEST; BOTTOM ROWS: COMPONENT-WISE GAIN OVER BEST PUBLISHED BASELINE (†/‡).

System	K	Dev	Mondial	Enterprise	ISWC	NPD	Held-out mean	Ten-block
Ontop†	–	.62/.48/.54	.56/.28/.51‡	.00/.00/**.51**	.53/.11/.39	.85/**.85**/.48‡	.48/.31/.47‡	.56/.42/.51‡
RDB2Onto†	–	.64/.64/.56	.54/.31/.67‡	.00/.00/.50	.67/.62/.44	.70/.81/.49‡	.48/.44/.52‡	.57/.56/.54‡
Gemini 2.5 Pro†	–	.77/.70/.80	.68/.50/**.84**	.02/**.10**/.34	.65/**.68/.56**	.30/.16/.08	.41/.36/.46	.63/.56/.66
DM (W3C baseline)	1	.65/.50/.60	.56/.28/.80	.00/.00/**.51**	.53/.21/.47	.87/**.85/.51**	.49/.33/.57	.58/.43/.59
DET (ours, 0 calls)	1	.86/.82/.84	.65/.55/.79	.00/.00/**.51**	.62/.62/.51	.75/.79/**.51**	.50/.49/**.58**	.72/.69/.73
DET-V2 (ours, 0 calls)	1	.86/.82/.84	**.70**/.55/.80	.00/.00/**.51**	.67/.62/.51	.84/.79/**.51**	.55/.49/**.58**	.74/.69/.73
HYBRID2 (ours)	1	**.94/.91/.94**	.64/.55/.79	.00/.00/**.51**	.53/.62/.52	.72/.77/.50	.47/.49/**.58**	**.75/.74/.80**
WHOLE-DB DeepSeek	5	.71/.70/.81	.69/**.57**/.79	.00/.00/.00	.67/.62/.51	.62/.57/.31	.49/.44/.40	.63/.60/.64
WHOLE-DB Kimi-K3	3	.79/.70/.79	**.70**/.56/**.84**	.26/.09/.43	**.73**/.67/.52	.80/**.85**/.49	**.62/.54**/.57	.72/.64/.71
PER-TABLE DeepSeek	10	.73/.68/.80	.69/.55/.78	.09/.03/.33	.55/.62/.51	.85/.76/.41	.55/.49/.51	.66/.61/.69
PER-TABLE Qwen3.6-35B-A3B	6	.70/.64/.80	.56/.45/.71	.10/.07/.49	.53/.40/.51	.83/.68/**.51**	.51/.40/.55	.62/.54/.70
PER-TABLE Qwen3.8-Max	6	.75/.68/.77	.53/.42/.72	.09/.08/.50	.60/.51/.51	.86/.80/.50	.52/.45/.56	.66/.59/.68
PER-TABLE Kimi-K3	6	.78/.64/.82	.62/.49/.75	**.34/.10**/.44	.57/.67/.51	**.88**/.82/.50	.60/.52/.55	.71/.59/.71
DET-V2 *gain vs. best pub.*	–	+.09/+.12/+.04	–	–	–	–	+.07/+.05/+.06	+.11/+.13/+.07
Best-ours gain vs. best pub.	–	+.17/+.21/+.14	–	–	–	–	+.14/+.10/+.06	+.12/+.18/+.14

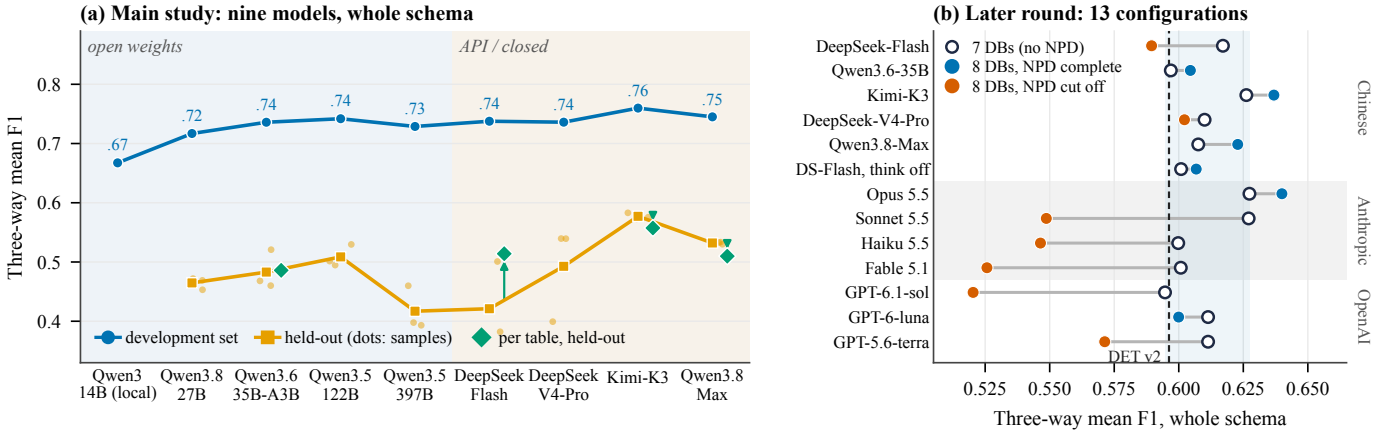


Fig. 3. Model choice is second-order once input and output are complete. (a) Main study, whole-schema generation with corrected prompts: three-way mean F1 on Dev and four Burr held-out schemas (line: $K = 3$ mean; dots: samples; green diamonds/arrows: per-table shift; DeepSeek held-out mean is .42 over 3 samples vs. .44 over 5 in Table I). (b) Extension round: 13 whole-schema configurations across seven non-NPD schemas and all eight (blue: all three NPD samples complete; dashed line: DET-V2).

B. *RQ2: Completeness Census, Decomposition, and Guarding*

1) Input-Completeness Diagnosis vs. Output Truncation: Initial runs showed every whole-schema model collapsing on 70-table NPD (0.05–0.22 class F1, crosses in Fig. 4a) while PER-TABLE reached 0.85—seemingly proving that large schemas require decomposition. However, a pre-registered stress test over FK-connected sub-schemas of 10, 20, and 40 tables [10, Sec. S2] showed flat accuracy (0.71–0.74, Fig. 4c), isolating the drop to a DDL-filter bug above 60k characters: testing statement keywords before stripping `pg_dump` comment headers silently dropped 66 of NPD's 70 `CREATE` statements [10, Fig. S7a]. With complete DDL, every finished single call achieves 0.77–0.87 class F1 on NPD, matching PER-TABLE (except Qwen3.8-27B template errors at 0.32, Fig. 4a). What remains on large schemas is *output* truncation: across all 9,673 whole-schema outputs in our study, incomplete mappings rise from 0.2% on 1–5 tables to 27% on 51–100 tables (Fig. 4b). The initial PER-TABLE builder also omitted `CREATE` bodies (keeping constraint `ALTER`s and sample rows) on 66 NPD tables, lowering schema-only NPD attribute F1 from 0.49 to 0.15 (Table II, bottom row).

2) Multi-Database Evaluation Across Eight Schemas: With corrected whole-schema prompts across all eight data-bearing databases, our pre-registered hypothesis that PER-TABLE (35B) outperforms WHOLE-DB (Kimi-K3) is refuted ($-.053$ class, $-.081$ relation F1; Holm $p = .75, .53$). Within-model differences (PER-TABLE minus WHOLE-DB) are similarly small [10, Fig. S4]: $+.030/+.026/+.014$ for DeepSeek, $-.044/-.028/+.013$ for 35B (class Holm $p = .013$), and $-.015/-.023/-.017$ for Kimi-K3, staying within $[-.13, +.08]$ everywhere except when PER-TABLE repairs truncated single calls (e.g., DeepSeek $+.28$ on NPD; Claude

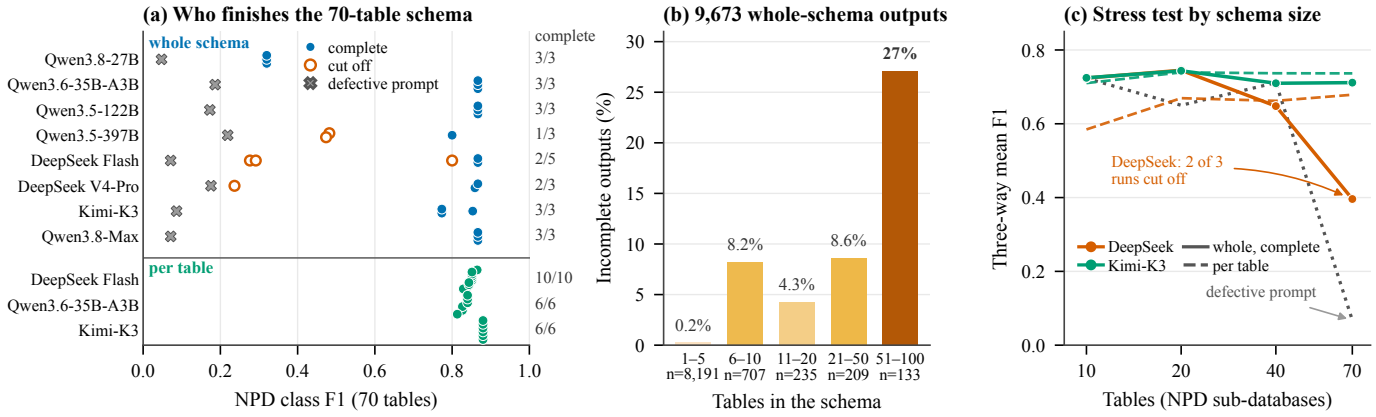


Fig. 4. Input and output completeness govern large-schema performance. (a) NPD (70 tables) class F1 per sample (hollow: $< 90\%$ table coverage; crosses: defective whole-schema prompt; PER-TABLE uses initial builder). (b) Incomplete-output rate across all 9,673 whole-schema runs by schema size. (c) Pre-registered sub-schema stress test (10–40 tables) vs. full 70-table NPD (dotted: defective prompt).

Sonnet/Haiku and GPT-6.1-sol 0.00–0.17 $\rightarrow$ 0.69–0.73).

3) Paired Complete-DDL Experiment and Completeness Guard: To compare calling regimes without input-completeness confounds, we ran a controlled complete-DDL schema-only experiment (fixed request and analysis plans prior to scoring, not pre-registered): Qwen3.8-Max, DeepSeek-V4.1-Flash, and Kimi-K3 (thinking off, temperature 0, three runs each) mapped all eight databases whole-schema and per-table (target table plus declared FK neighbours or top-3 lexically similar tables; 2,049 normal completions; zero scorer discrepancy across all 150 whole-DB mappings [10, Sec. S3]).

Table II, Fig. 5, and Fig. 6 establish four results. *First, full per-table decomposition yields no average accuracy gain at* $11.6\times$ *input tokens.* Across eight databases, PER-TABLE minus WHOLE-DB averages 0.00, -0.03, and -0.01 F1 for Qwen, DeepSeek, and Kimi-K3 (95% bootstrap CIs $[-0.03, +0.03]$, $[-0.09, +0.03]$, $[-0.04, +0.01]$), while consuming $11.6\times$ total input tokens (5.8–$22\times$ per database, reflecting repeated neighbour context and per-table prompts) and $1.34\times$ output tokens. On 70-table NPD, Kimi-K3's $+.05$ mean stems from one single call omitting classes on 15 tables (its two complete calls score 0.74 vs. 0.72 per table; similarly, Claude Opus 5.5 finished every schema in one call and scored 0.02 lower per table [10, Sec. S4]).

Second, local wins and losses reflect schema-open convention choices rather than reasoning depth (Fig. 5b). On Mondial, Qwen and DeepSeek single calls expose 100–115 key/FK columns as data properties (which gold excludes), whereas PER-TABLE exposes 20–55 ($+0.32$ attribute precision); conversely, PER-TABLE over-generates undeclared `publisher` relations on BSBM (16–23 vs. 11 gold), turns ISWC link tables into classes for DeepSeek, and lowers Kimi-K3's Mondial class/relation precision by 0.34/0.22. A diagnostic post-hoc filter stripping data properties on URI-template/join columns from single-call outputs lifts Mondial attribute F1 from 0.52/0.51 to 0.68/0.66 at zero calls (≤ 0.015 shift elsewhere), and adding one key-column rule to the single-call prompt (72 calls [10, Sec. S3], Fig. 5c) closes half to two-thirds of the Mondial gap, leaving PER-TABLE at or below instructed single calls across all three models.

Third, our selective completeness guard retains single-call accuracy at $1.12\times$ *input tokens—roughly one-tenth the cost of* PER-TABLE (Fig. 6a). Evaluated on the paired archive, the unmentioned-table guard triggered on only 4 of 72 single calls (19 tables total, 18 from misspelt DeepSeek Mondial table names), staying within 0.002 F1 of WHOLE-DB across all three models and scoring 0.003–0.029 *higher* than full PER-TABLE at $1.12\times$ vs. $11.6\times$ input tokens. A stricter guard requiring a class or attribute per table recovers Kimi-K3's partial NPD run ($0.55 \rightarrow 0.65$) at $2.1\times$ tokens, but re-queries link tables already modelled as relations and converts them into classes (-0.03 for DeepSeek; -0.007 even after exempting all-key tables, Fig. 6b), confirming that minimal unmentioned-table guarding best preserves global modelling coherence.

Fourth, test-time reasoning alters convention choices and risks output-budget exhaustion (Fig. 6c). Enabling reasoning (48 single calls) raises Qwen $+0.24$ on Mondial (suppressing key attributes) but drops -0.12 on NPD (emitting 0–186 vs. $\sim$ 830 attributes), netting $+0.011$ over eight databases at $2.5\times$ output tokens and $4.1\times$ latency. All three Kimi-K3 NPD calls spent 22k–26k reasoning tokens and hit the 65,536-token ceiling; doubling the ceiling restores 0.73 F1 (-0.003 eight-DB delta), and our table-level guard automatically catches such budget-truncated outputs [10, Sec. S3].

C. *RQ1: Data-Driven Rules vs. Declared-Key Removal*

Zero-call DET-V2 *beats published rule baselines and rivals frontier LLMs when keys exist, while* `@stripped` *isolates LLM necessity:* Without a single model call, DET reaches .86/.82/.84 on Dev—beating Ontop (.62/.48/.54), RDB2Onto (.64/.64/.56), published Gemini 2.5 Pro (.77/.70/.80), and every un-tuned WHOLE-DB checklist run (Table I). With pre-registered refinements, DET-V2 achieves .55/.49/.58 on Burr's held-out schemas ($+.07/+.18/+.11$ over Ontop and $+.07/+.05/+.06$ over RDB2Onto) and 0.611 across the eight data-bearing databases—beating DM (0.542) by $+0.069$, matching PER-TABLE LLM pipelines (0.585–0.619), and trailing the strongest WHOLE-DB LLM

TABLE II
PAIRED COMPLETE-DDL SCHEMA-ONLY EXPERIMENT: MAPPING-BASED F1 (CLASS/RELATION/ATTRIBUTE, 3-RUN MEAN) AND PER-TABLE MINUS WHOLE-SCHEMA THREE-WAY MEAN (Δ). TABLE COUNTS IN PARENTHESES; LAST ROW SHOWS DEFECTIVE-DDL IMPACT ON NPD PER-TABLE.

	Qwen3.8-Max			DeepSeek-V4.1-Flash			Kimi-K3		
Database	Whole	Per table	Δ	Whole	Per table	Δ	Whole	Per table	Δ
Bgee (6)	.57/.63/.57	.57/.63/.58	.00	.57/.63/.60	.57/.63/.60	.00	.57/.63/.60	.57/.63/.58	−.01
ODH (8)	.35/.45/.78	.35/.39/.75	−.03	.35/.45/.79	.35/.41/.76	−.03	.35/.45/.81	.35/.45/.77	−.01
ISWC (9)	.73/.62/.52	.69/.61/.51	−.02	.73/.62/.53	.53/.43/.51	−.13	.73/.67/.52	.73/.67/.51	.00
BSBM (10)	1.00/.76/.95	1.00/.57/.96	−.06	1.00/.76/.95	.84/.36/.95	−.19	1.00/.76/.95	1.00/.66/.95	−.04
GTFS (10)	.87/.83/.98	.87/.86/.97	+.01	.87/.77/.99	.79/.84/.96	−.01	.87/.79/.99	.92/.78/.98	+.01
LUBM (14)	.07/.26/.11	.08/.24/.11	.00	.08/.26/.11	.07/.24/.11	−.01	.08/.26/.11	.08/.26/.10	.00
Mondial (46)	.56/.26/.52	.58/.38/.69	+.11	.56/.31/.51	.56/.44/.80	+.14	.68/.55/.84	.58/.45/.78	−.09
NPD (70)	.87/.83/.50	.87/.77/.49	−.02	.87/.72/.50	.80/.76/.49	−.01	.79/.75/.49	.86/.82/.50	+.05
Mean (8 DBs)	.63/.58/.62	.63/.56/.63	.00	.63/.57/.62	.57/.51/.65	−.03	.63/.61/.66	.64/.59/.65	−.01
NPD, defective builder	–	.87/.84/.15	−.09	–	.84/.77/.15	−.10	–	–	–

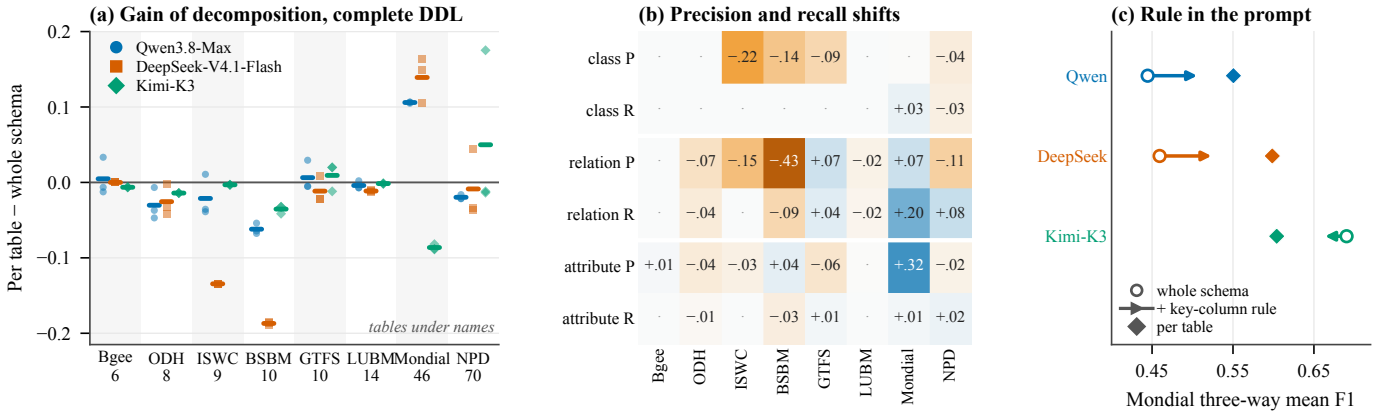


Fig. 5. Paired complete-DDL experiment. (a) Per-table minus whole-schema three-way F1 per database (bars: 3-run mean; markers: runs). (b) Precision/recall shift by element type (Qwen and DeepSeek mean; dots: $|\Delta| < .005$). (c) Mondial three-way F1 of single call, single call with key-column prompt rule (arrow), and per-table generation.

by just 0.026–0.029 (Fig. 1). Specifically, DET-V2's type-discriminator rule—which never fires on Dev—boosts Burr held-out class F1 from .671 to .735 (+.064, Holm $p = .002$, 100% precision on withheld classes), whereas parent-to-child orientation helped Dev (.691 → .746) but hurt held-out extent F1 (.260 → .189) due to inverted gold conventions and was reverted [10, Table S2]; HYBRID2 leads Dev (.94/.91/.94) and Ten-block (.75/.74/.80) before reverting to DET's held-out level.

Removing all declared PK, FK, and unique constraints (`@stripped`, pre-registered [10, Sec. S4]) reveals the exact boundary where LLMs add irreplaceable semantic value (Fig. 7a): DM collapses to 0.00 relation F1 and DET-V2 drops from 0.56 to 0.35 (as pure profiling recovers 94%/82% of PKs/FKs on Dev but 48%/36% on complex held-out schemas), whereas four LLM configurations retain 91–100% of their relation F1 (0.53–0.60), outperforming DET-V2 by +0.18 to +0.25 (Holm $p \leq .005$). Meanwhile, pre-registered synonym renaming across all eight databases shifts LLM scores by only −0.016 to +0.016 (Holm $p \geq .64$, Fig. 7b), ruling out benchmark-name memorization.

D. *RQ3: Convention Overfitting and Cross-Distribution Risk Control*

Explicitly encoding Dev's modelling conventions into the prompt boosts DeepSeek to .95/.94/.95 on Dev (PER-TABLE: .94), yet collapses by −0.44 to 0.50 on Burr's held-out schemas (Fig. 7c), mirroring positive Dev–held-out gaps across all ten systems on the five added databases. Similarly, CERT satisfies finite-sample FDP control at every target α under leave-one-category-out Dev calibration, but breaks when those thresholds are transferred to real databases (Fig. 7d): at $\alpha = .20$, realised FDP reaches .315/.205/.258 (and .317/.232/.222 with corrected WHOLE-DB pools; pooled class FDP .275 at $\alpha = .10$ on the five added databases) [10, Sec. S7]. Because real-database gold conventions diverge systematically from micro-benchmark conventions on decisions where *all* models and rules agree, cross-source agreement violates cross-distribution exchangeability—proving that valid FDP guarantees require real-schema calibration, even though CERT's agreement score still cleanly ranks uncertainty (50% precision on 161 suggested vs. 8% on 89 abstained elements).

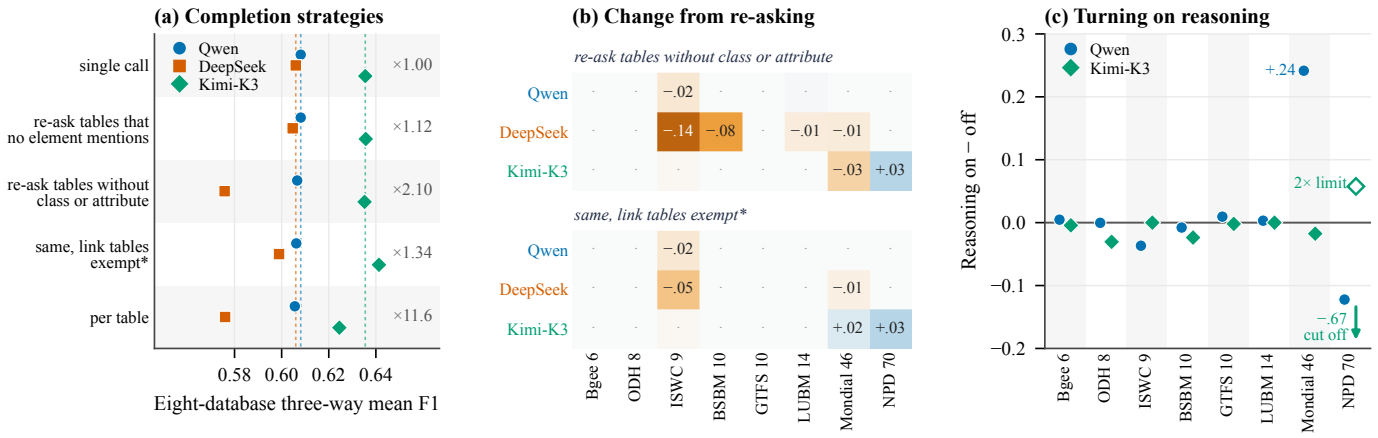


Fig. 6. Selective completeness guarding vs. full per-table decomposition on the paired experiment. (a) Eight-database three-way F1 and relative input tokens (1.12× for our primary guard vs. 11.6× for PER-TABLE; dotted: single call; *: post-hoc all-key exemption). (b) Per-database F1 delta of stricter guards vs. single call. (c) Impact of enabling test-time reasoning per database (hollow marker: Kimi-K3 NPD with doubled output-token ceiling).

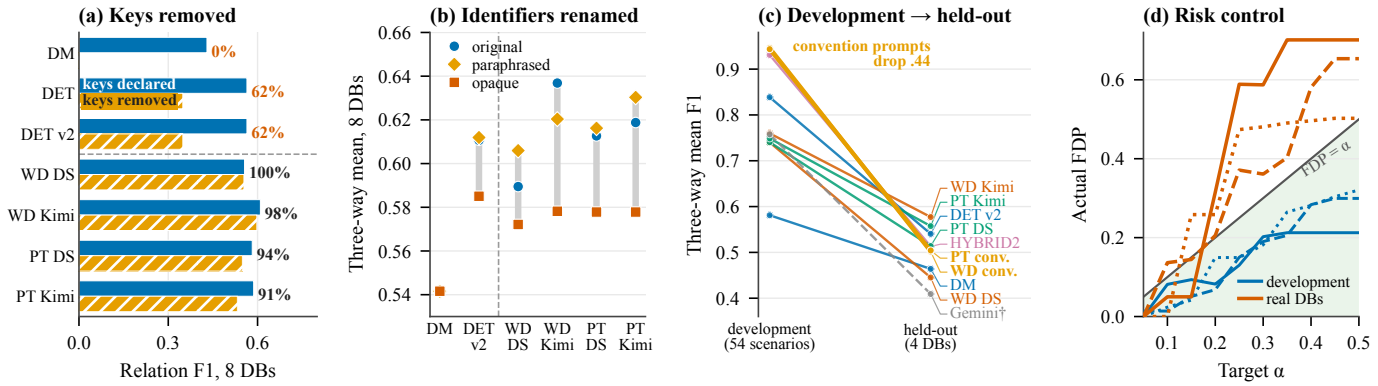


Fig. 7. Constraint-removal boundary, memorization test, and convention transfer. (a) Eight-database relation F1 with declared keys vs. `@stripped` (all keys removed; percentages show retention). (b) Three-way F1 under original, synonym-paraphrased, and opaque identifiers. (c) Dev vs. Burr held-out F1 showing convention-prompt overfitting (conv.: 0.95 → 0.50). (d) Realised FDP of CERT vs. target α on Dev (leave-one-category-out) and four real databases.

*E. **RQ4**: Extent-Based Evaluation Reveals Convention Sensitivity*

Evaluating mapped instance tuples via our extent metric (Fig. 8a) leaves class F1 stable but lowers held-out relation F1 by 0.37–0.46 for rule systems vs. $\sim$ 0.12 for LLMs: 45–93 rule-generated relations per schema invert the gold subject–object convention, which Burr's direction-agnostic join comparison counts as exact matches (while Burr's class-agnostic attribute key shifts attributes oppositely). Thus, evaluation semantics directly invert whether rules or LLMs lead on relations.

F. Scalability, Cost–Quality Frontier, and Query Evaluation

On scalability and cost (Fig. 8c), DuckDB profiling plus DET-V2 processes TPC-H [51] sf30 (2.6×10^8 rows) end to end in 14 minutes (26.8 GB peak RAM) with stable key-free recovery [10, Sec. S2], defining the Pareto cost–quality frontier alongside single-call Kimi-K3 (and our 1.12×-token guarded call). On RODI [8] (Fig. 8b), all six systems cluster at 0.35–0.40 query F1 from the database alone because unaligned target IRIs fail SPARQL matching; supplying target vocabulary lifts LLMs to 0.53–0.61, confirming RODI primarily measures ontology alignment rather than schema-to-ontology learning [10, Sec. S2].

VII. DISCUSSION

Why completeness and conventions dominate: Mapping accuracy is upper-bounded by input and output completeness: tables omitted by prompt filters or truncated before output closure lose all downstream elements. Once outputs are complete, residual variance concentrates on schema-underdetermined conventions—whether key columns double as attributes, whether relations point child-to-parent or parent-to-child, and whether link tables form classes. Because LLMs across parameter scales resolve these ambiguities similarly, model choice is second-order, cross-model agreement reflects shared priors rather than independent verification, convention-tuned prompts fail across benchmarks, and complete-by-construction profiling rules (DET-V2) closely rival frontier LLMs whenever keys are declared.

Actionable guidelines for practitioners: First, when constraints are declared, run zero-call data-profiling rules (DET-V2) as the default baseline or backbone (0.611 vs. 0.637–0.640 best LLM); reserve LLM inference for schemas lacking declared keys (`@stripped`, +0.18 to +0.25 relation F1).

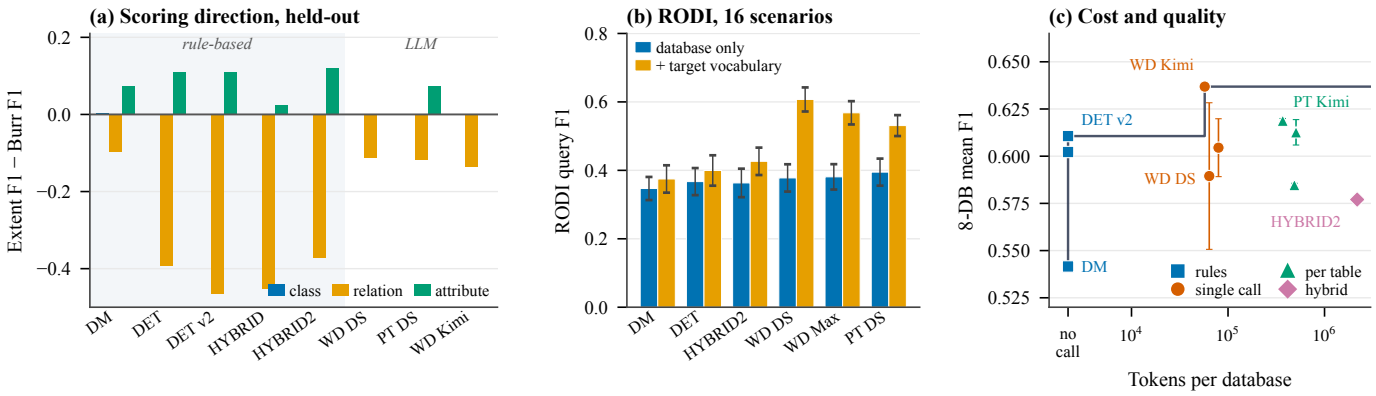


Fig. 8. Extent evaluation, query benchmarking, and cost–quality Pareto frontier. (a) Instance-extent F1 minus Burr syntactic F1 on held-out schemas ($\tau = 1$). (b) RODI query F1 across 16 scenarios (mean ± s.e.) without vs. with target vocabulary. (c) Eight-database three-way F1 vs. total tokens per database (Mondial, ISWC, NPD mean; line: Pareto frontier).

Second, replace 11.6×-token per-table decomposition with a single whole-schema call protected by our *selective completeness guard* (1.12× tokens, within 0.002 F1 of whole-schema), audit prompts to verify every table's `CREATE` statement is present, and allocate separate output budget when test-time reasoning is enabled. Third, align convention choices explicitly (e.g., filtering key-column attributes) and calibrate CERT risk bounds on real schemas rather than micro-benchmarks, using agreement scores to prioritise human review.

Implications for benchmark design: Syntactic evaluators like Burr advance beyond lexical matching but should explicitly score (or parameterise) relation direction, bind property equivalence to subject/object classes (fixing 14 misattached LUBM entries left syntactic scores unchanged), and audit gold mappings via instance extents (which caught 12 empty-extent gold elements). Future benchmarks also require versioned constraints and multi-dozen-table real schemas.

Threats to validity and limitations: Leakage and experimental tracks. Because DET and HYBRID2 were developed on the 54 micro-scenarios, all real-schema claims rest on sealed held-out evaluation and pre-registered updates. The main study included sample rows and the initial NPD per-table builder, whereas the 2,049-call paired experiment used complete schema-only prompts across three models ($K = 3$) under fixed, non-pre-registered plans (and its per-table ownership prompt differs from the single-call checklist, though adding a key-column prompt rule leaves eight-database conclusions unchanged); HYBRID2 and CERT's per-table sources were not rerun under schema-only prompts, and the post-hoc key-attribute filter is a diagnostic control rather than an independently validated deployment rule. *Sample size and baselines.* Eight data-bearing held-out databases (three generated, two ≥ 46 tables) yield coarse bootstrap power, so scale claims rest on the 9,673-output truncation mechanism. Gemini 2.5 Pro figures are Burr's published pre-correction values, whereas our runs and Mondial/NPD Ontop and RDB2Onto rows use the constraint-corrected release without re-running external baselines. *Gold conversions and API drift.* Converted mappings drop 10 of 342 inexpressible entries and approximate 47; instance execution verifies agreement on current data snapshots only [10, Sec. S5]. Commercial API endpoints may evolve despite logged dates and archived responses, and profiling rules cannot infer semantics hidden in boolean flags or composite string templates.

Reproducibility: Offline container replay of 4,735 scoring and 9,673 coverage tasks reproduces all eight core result tables byte for byte, and all 2,049 paired-experiment requests, streamed responses, and token logs were hash-sealed prior to gold scoring [10, Sec. S3]. Bgee data (CC0) is publicly cleared; our standalone scorer, extent evaluator, preregistrations, outputs, and five converted mappings will be released upon licence clearance.

VIII. CONCLUSION

We presented a unified dual-route framework and multi-benchmark investigation of relational-to-ontology learning across 25 configurations, 16 LLMs, Burr, RODI, and five newly converted, execution-verified databases. Our results show that real-database mapping quality is governed primarily by input/output completeness and modelling conventions rather than LLM scale (0.032 F1 spread across 13 whole-schema configurations on seven held-out databases). On the methodological front, our zero-call data-profiling engine DET-v2 substantially outperforms Ontop and RDB2Onto (+0.05 to +0.27 F1) and comes within 0.03 of the strongest frontier LLM when keys are declared, while key-removal experiments isolate the +0.18 to +0.25 relation-F1 regime where LLMs are indispensable. Under complete DDL, our selective *completeness guard* matches whole-schema accuracy (within 0.002 F1) at 1.12× input tokens—eliminating the 11.6× token overhead of full per-table decomposition, which brings no average accuracy gain (−0.03 to 0.00 F1). Finally, our cross-source controller CERT and instance-level *extent metric* expose why micro-benchmark convention prompts (0.95 → 0.50) and FDP bounds fail under real-schema convention shifts and reveal a 0.37–0.46 relation-F1 gap hidden by direction-agnostic scoring, providing rigorous tools and baselines for reliable ontology learning.

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